The Fundamental Theorem of calculus

Theorem

If f in continuous on $[a,b]$ and if F is any anti-derivative of f, that is $F' = f$, then

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

Proof

Divide the interval $[a,b]$ in n subintervals with equal length $\triangle x = \dfrac{b-a}{n}$.

Let $x_0 = a < x_1 < \ldots. < x_{n-1} < x_n = b$ be the endpoints of the subintervals. ....(**A**)

Let F be an anti-derivative of f, that is F is a function such that $F' = f$. (**B**)

We apply the Mean Value Theorem to F on $[x_{i-1}, x_i]$:

- F is differentiable on (a,b) (if $F' = f$, then the derivative exists)
  so F is differentiable on (x_{i-1}, x_i)
- F is continuous on $[a,b]$ (because F is differentiable).
  so F is continuous on $[x_{i-1}, x_i]$

So there exists a number $c_i \in (x_{i-1}, x_i)$ such that

$$\frac{F(x_i) - F(x_{i-1})}{x_i - x_{i-1}} = F'(c_i)$$

$$= f(c_i) \qquad \text{(from B)} \qquad (\mathbf{C})$$

Take the point c_i as the sample point in every interval.

Because f is continuous on $[a,b]$, then f is integrable on $[a,b]$. It follows that

$$\int_a^b f(x)\,dx$$

$$= \lim_{n\to\infty} \sum_{i=1}^{n} f(c_i)\Delta x$$

$$= \lim_{n\to\infty} \sum_{i=1}^{n} \frac{F(x_i) - F(x_{i-1})}{x_i - x_{i-1}}\Delta x \qquad \text{From the Mean Value Theorem (} \mathbf{C} \text{ above)}$$

$$= \lim_{n\to\infty} \sum_{i=1}^{n} \frac{F(x_i) - F(x_{i-1})}{x_i - x_{i-1}}(x_i - x_{i-1})$$

$$= \lim_{n\to\infty} \sum_{i=1}^{n} F(x_i) - F(x_{i-1})$$

$$= \lim_{n\to\infty} \big([F(x_1) - F(x_0)] + [F(x_2) - F(x_1)] + [F(x_3) - F(x_2)] + \ldots + [F(x_{n-1}) - F(x_{n-2})] + [F(x_n) - F(x_{n-1})]$$

$$= \lim_{n\to\infty} \big([F(x_n) - F(x_0)]$$

$$= \lim_{n\to\infty} \big([F(b) - F(a)] \qquad \text{(From } (\mathbf{A})\text{)}$$

$$= F(b) - F(a)$$